

PCA9685 16-Channel 12-Bit PWM Servo Driver

Sunfounder

Introduction

PCA9685 16-Channel 12-Bit PWM Servo Driver can help a lot when there are insufficient PWM output pins for the MCU in your project. The only condition is that the main control chip supports I2C communication, which means enabling the communication between the chip and PCA9685, so as to control multiple servos simultaneously.

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Key Features:

Autonomous
Scalable
High-Precision

01

I2C PWM Control: built-in clock with 16 free-running PWM outputs, no continuous MCU signal required.

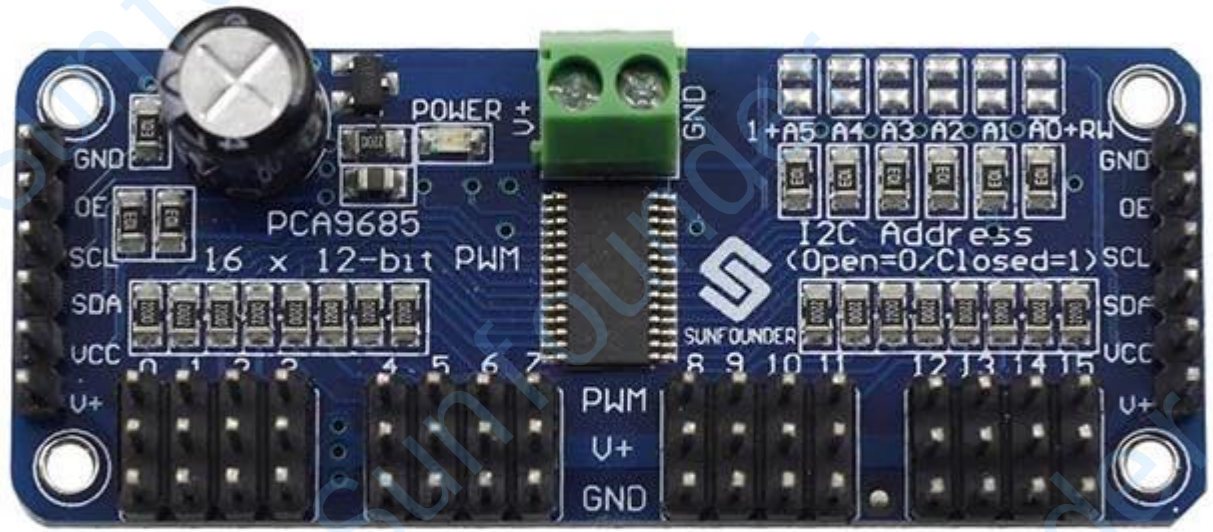
02

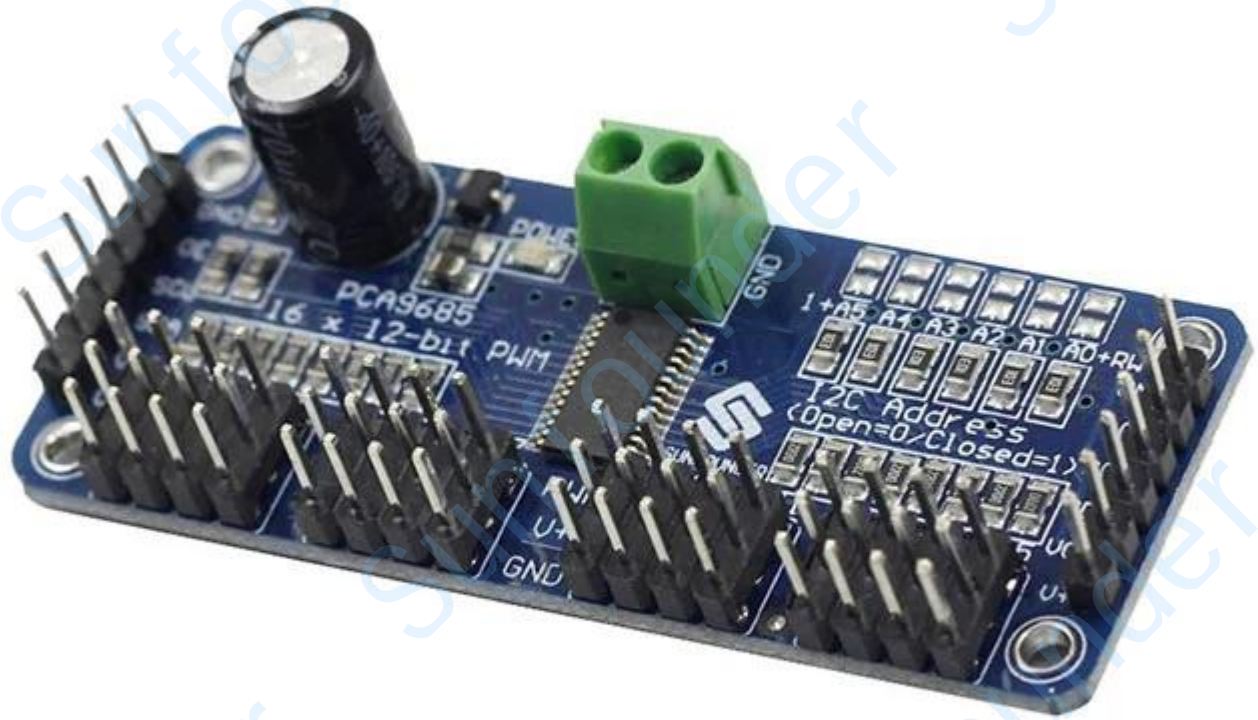
High Expandability & Compatibility: 5V compliant, works with 3.3V logic, supports chaining up to 992 outputs.

03

High Precision & Servo Support: 12-bit resolution with 16 servo-capable channels for accurate control.

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POWER
GND
OE
SCL
SDA
VCC
V+
PCA9685
16 x 12-bit PWM
0 1 2 3 4 5 6 7
PWM
V+
GND
8 9 10 11 12 13 14 15
SUNFOUNDER
I2C Address
(Open=0/Closed=1)
1+A5 A4 A3 A2 A1 A0+RW


PCA9685
16 x 12-bit PWM
GND
I2C Address
(Open=0/Closed=1)


PCA9685 16-Channel 12-Bit PWM Driver
Use V+ for servo supply
VCC: 3-5V
Logic: 3-5V
V+: 6V Max
Freq.: 40-1000Hz
Terminal block is reverse polarity protected. Side breakout pins are not!

Package List

- PCA9685 16-Channel 12-Bit PWM Servo Drive x 1

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More Info:

PCA9685 16-Channel 12-Bit PWM Servo Driver



Thank you for your time and attention.

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